

- A/Prof Tay Seng Chuan
Email: pvotaysc@nus.edu.sg

Have you ever thought of

- how to calculate the electric potential energy of a collection of charges.
- the meaning and significance of electric potential.
- how to calculate the electric potential that a collection of charges produces at a point in space.
- how to use equipotential surfaces to visualize how the electric potential varies in space.
- how to use electric potential to calculate the electric field.

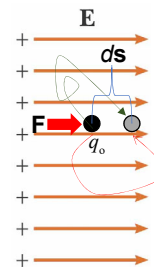


A few slides in this chapter come from Chapter-23 of Young's textbook.

1

Electric Potential Energy

- When a test charge q_o is placed in a (positive) electric field, it experiences a force $\mathbf{F} = q_o \mathbf{E}$. So the test charge will start to move away and accelerate on a path, and will gain kinetic energy due to the acceleration. Let $d\mathbf{s}$ be the small displacement.
- The force on the test charge in electric field is conservative. So only the start point and the end point of the path are concerned. The pattern of the trajectory, i.e., the path, does **not** affect our formulation.
- $d\mathbf{s}$ in the figure is an infinitesimal displacement vector that is oriented tangentially to a path through space. The **work** done by the electric field on the test charge is $\mathbf{F} \cdot d\mathbf{s} = F \times ds \times \cos 0^\circ$



dot product

$$= (q_o \mathbf{E}) \cdot d\mathbf{s}$$

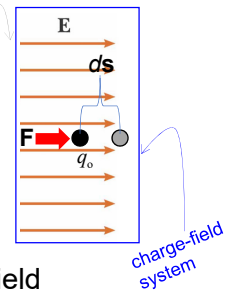
$$E = \frac{F}{q_o}$$

$$\rightarrow F = q_o E$$

2

Electric Potential Energy

source charge

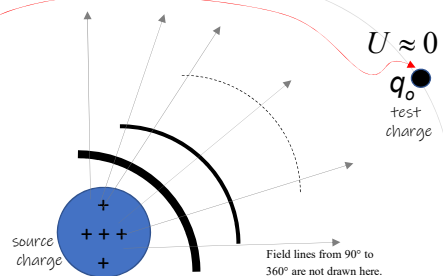


- The source charge produces the electric field.
- The charge-field system refers to the test charge q_0 and the field.
- As work is done by the field to move the test charge q_0 , the potential energy of the charge-field system is **now smaller** at the new location, and the **change** in potential energy of charge-field system is $\Delta U = -(q_0 \mathbf{E}) \cdot d\mathbf{s}$. The change of potential energy is negative because a portion of the initial potential energy in the charge-field system has been used to accelerate the test charge.
- For a finite displacement from A to B, the change in potential energy in the charge-field system is $\Delta U_{(A \rightarrow B)} = U_B - U_A = -q_0 \int_A^B \mathbf{E} \cdot d\mathbf{s}$ i.e., if q_0 gains faster speed (increase in kinetic energy), the charge-field system at the new location will have smaller potential energy.
- Because the force $q_0 \mathbf{E}$ is conservative, the line integral (or path integral) does not depend on the path taken by the test charge q_0

3

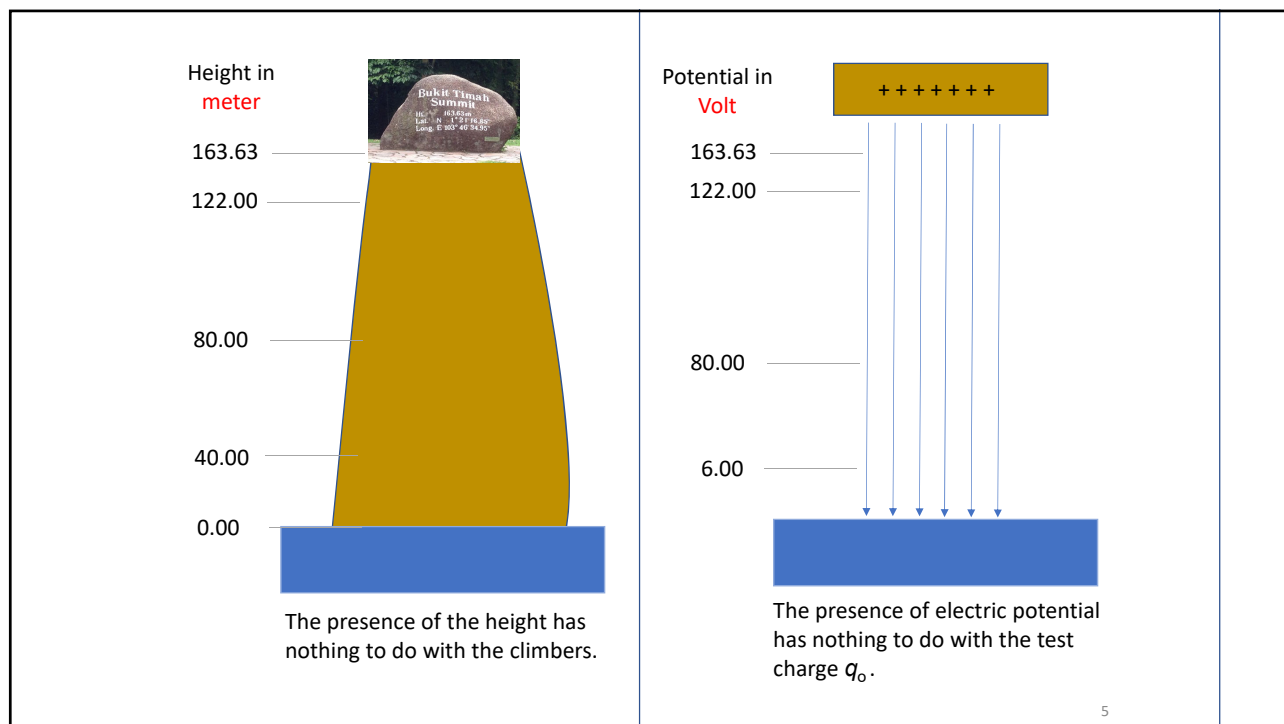
Electric Potential

- For the given location of the test charge q_0 in the field, the test charge in the charge-field system has a potential energy U relative to a particular configuration of the system that is defined as $U = 0$.



- The location for **zero** potential of the charge-field system can be **very far** away from the source charge, and when a test charge is placed there it will not acquire any potential energy.
- The **electric potential (V)** of the charge-field system at a particular location refers to the **potential energy per unit (test) charge** there, i.e., $V = \frac{U}{q_0}$
- At every point in an electric field, the electric potential there has a specific value
- The sign of the electric potential depends on the sign of the source charge
- The magnitude of electric potential is smaller if the location there is farther away from the source charge
- The electric potential is independent of the value of q_0 (because the numerator also has a q_0) (The height of Bukit Timah Hill is not affected by the heights of climbers!!)

4



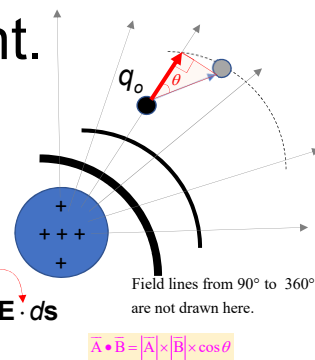
Electric Potential, cont.

- The electric potential is a scalar quantity since **energy and charge are scalar**
- As a charged particle moves in an electric field, it will experience a **change** in potential denoted as

$$\Delta V = \frac{\Delta U}{q_o} = \int_A^B \frac{(\mathbf{F} \cdot d\mathbf{s})}{q_o} = - \int_A^B \left(\frac{\mathbf{F}}{q_o} \right) \cdot d\mathbf{s} = - \int_A^B \mathbf{E} \cdot d\mathbf{s}$$

change in potential energy ΔU
change in potential ΔV
charge q_o

dot product



- Electric potential is a scalar characteristic of an electric field, independent of any charges that may be subsequently placed in the field. So the existence (or non-existence) of q_o is not important in our formulation (You can map electric potential to the height of a mountain)
- We often take the value of the potential to be zero at some convenient point designated as infinity (very far from source charge) in the field
- The preferred convenient point for the zero potential does not affect the **difference** in potential between any two locations.

Work and Electric Potential

- Since the changes in potential is the changes in potential energy for one unit of charge, i.e., $\Delta V = \frac{\Delta U}{q_o}$

the work done **on** the charge will be $(q_o \times \Delta V)$, and this energy can be exhibited by a faster speed.

- The unit of potential is in volts (V), i.e., $1 \text{ V} = 1 \text{ J/C}$
 - V is a volt
 - It takes one joule of work to move a 1-coulomb of charge through a potential difference of 1 volt

Volt

$$\Delta V = \frac{\Delta U}{q_o} = \int_A^B \frac{-(F \cdot ds)}{q_o} = - \int_A^B \left(\frac{F}{q_o} \right) \cdot ds = - \int_A^B \mathbf{E} \cdot d\mathbf{s}$$

So $1 \frac{\text{N}}{\text{C}} = 1 \frac{\text{V}}{\text{m}}$

Meter
Newton per Coulomb

This indicates that we can interpret the magnitude of **electric field** as a measure of the magnitude for the rate of change of the electric potential with position. (Electric Field **E** can be treated as the negative of potential gradient i.e., $\mathbf{E} = -\frac{dV}{ds}$)

7

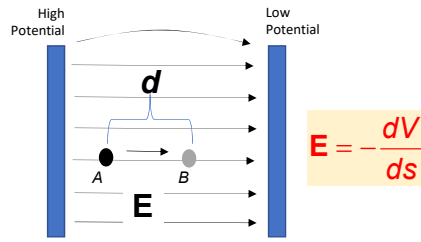
Electron-Volts (eV) *eV is energy!!*

- Another unit of energy (in Joule) that is commonly used in atomic and nuclear physics is the electron-volt
- One **electron-volt** is defined as the energy a charge-field system gains or loses when a charge of magnitude **e** (an electron or a proton) is moved through a potential difference of 1 volt
 - $1 \text{ eV} = 1.60 \times 10^{-19} \text{ coulomb} \times 1 \text{ volt} = 1.60 \times 10^{-19} \text{ J}$

8

Potential Difference in a Uniform Field

- The equations for electric potential can be simplified if the electric field is uniform. For uniform field $\mathbf{E}$, the electric field has a constant magnitude.



Electric field is the negative of the potential gradient with respect to displacement.

$$V_B - V_A = \Delta V = -\int_A^B \mathbf{E} \cdot d\mathbf{s} = -E \int_A^B ds = -Ed$$

The negative sign indicates that the electric potential at point B is lower than at point A, i.e., *work is done by the field to move the charge from location A to location B thus the potential in the field has become lower. But the charge after moving from point A to point B will gain faster speed!!*

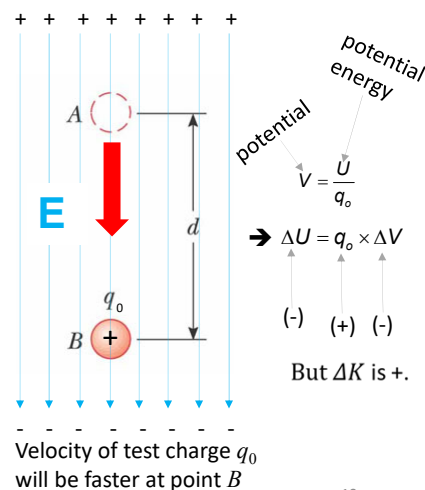
(A uniform electric field does not mean that the electric potential is also uniform!!)

9

Energy and the Direction of Electric Field

(Water flows from high ground to the low ground. Electric field directs from high potential to low potential)

- When the electric field $\mathbf{E}$ is directed downward from point A to point B, point B will be at a lower potential than point A
- So, when a positive test charge moves along the electric field from A to B, the charge-field system loses potential energy as the test charge gains kinetic energy



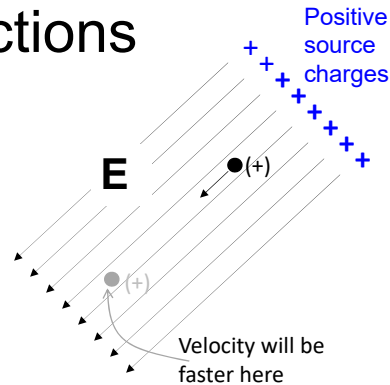
10

More About Directions

- A system consisting of a positive charge and an electric field **loses** electric potential energy when the positive charge moves in the direction of the field

✓ An electric field does work **on** a positive charge when the charge moves **in the direction of the electric field**

- The charged particle gains kinetic energy due to the loss of potential energy at the new location in the charge-field system
- ✓ This is an example of **Conservation of Energy**



11

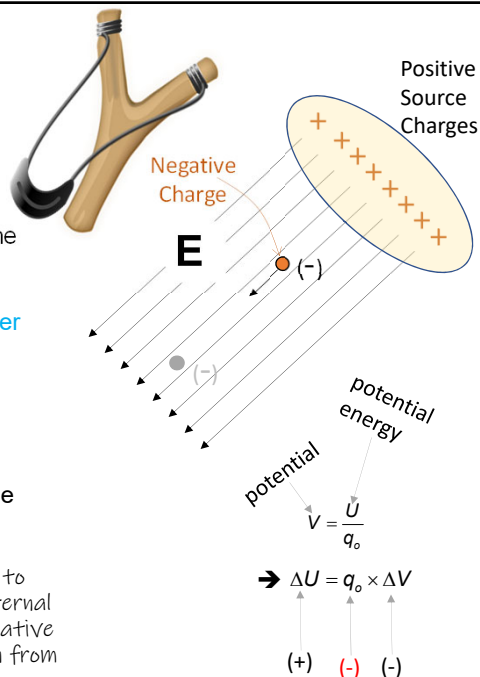
Directions, cont.

- If q_o is **negative**, then ΔU in q_o is positive when it is farther away from the positive source charge. (In this case you can consider the field line as rubber band!!)



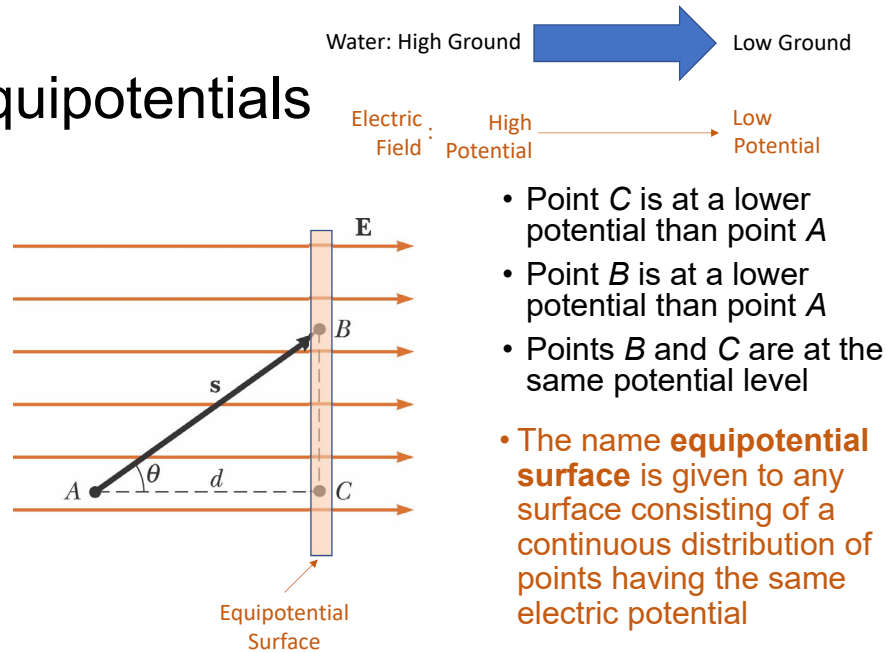
- A system consisting of a negative charge **gains** electric potential energy when the negative charge moves in the direction of the field

(because in order for a negative charge to move in the direction of the field, an external agent must do positive work on the negative test charge to overcome the attraction from the source.)



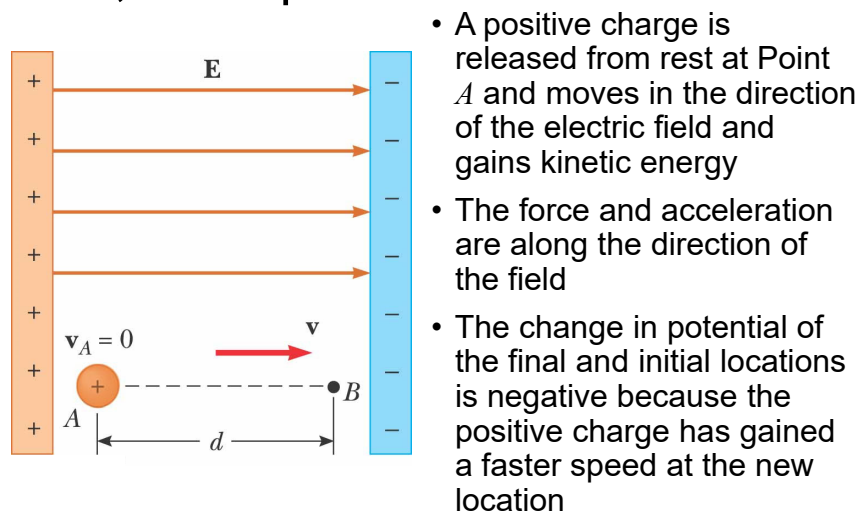
12

Equipotentials



13

Charged Particle in a Uniform Field, Example



14

Potential and Point Charges

(What if the electric field is not uniform?)

Potential Difference:

$$V_B - V_A = -\int_A^B \mathbf{E} \cdot d\mathbf{s}$$

But $\mathbf{E} \cdot d\mathbf{s} = \left(k \frac{q}{r^2} \right) \hat{\mathbf{r}} \cdot d\mathbf{s}$

$\hat{\mathbf{r}} \cdot d\mathbf{s} = 1 \times |d\mathbf{s}| \times \cos \theta = dr$

$$V_B - V_A = -\int_{r_A}^{r_B} \left(k \frac{q}{r^2} \right) dr = kq \left[\frac{1}{r} \right]_{r_A}^{r_B}$$

$$= kq \left[\frac{1}{r_B} - \frac{1}{r_A} \right]$$

$$\mathbf{E} = -\frac{dV}{ds}$$

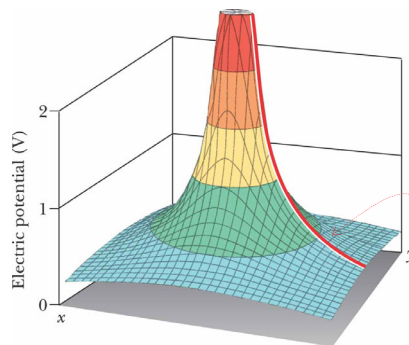
- A positive point charge produces a field directed radially outward
- The potential difference between points A and B will be $V_B - V_A = kq \left[\frac{1}{r_B} - \frac{1}{r_A} \right]$
- The electric potential is independent of the path between points A and B
- It is customary to choose a reference potential of $V = 0$ at $r_B = \infty$
- Then the potential at some point located at r from the source charge q is $V = k \frac{q}{r}$

Quantity of Source charge

Distance from Source charge

15

Electric Potential of a Point Charge



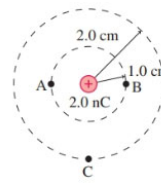
- The electric potential in the plane around a single positive point charge is shown in the figure
- The red line shows the $\frac{1}{r}$ hyperbola nature of the potential

The electric potential tends to 0 when the position from the source charge is too far (at infinity). The analogy is

16

Example. In the following figure the positions for points A, B and C are marked, and a $2 \times 10^{-9} \text{ C}$ charge is placed at the center.

- What is the electric **potential** at points A, B, and C?
- What is the electric **potential energy** of an electron at each of these points?
- What are the potential differences ΔV_{AB} and ΔV_{BC} ?



$$V = k \frac{q}{r}$$

Quantity of Source charge

Distance from Source charge

Answer:

The charge is a point charge.

- The electric potential from the point charge q at a distance r is

$$V = k \frac{q}{r} = (8.9875 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2) \left(\frac{2.0 \times 10^{-9} \text{ C}}{r} \right) = \frac{17.975 \text{ N} \cdot \text{m}^2 / \text{C}}{r}$$

For points A and B, $r = 0.01 \text{ m}$. Thus,

$$V_A = V_B = \frac{17.975 \text{ N} \cdot \text{m}^2 / \text{C}}{0.01 \text{ m}} = 1797.5 \frac{\text{N} \cdot \text{m}}{\text{C}} = 1797.5 \left(\frac{\text{V}}{\text{m}} \right) \text{m} = 1797.5 \text{ V}$$

For point C, $r = 0.02 \text{ m}$ and $V_C = \frac{17.975 \text{ N} \cdot \text{m}^2 / \text{C}}{0.02 \text{ m}} = 898.75 \text{ V}$

- The electric potential energy of a charge e at a point where the electric potential is V is $U = e \times V$. The expression for the potential in part (i) assumes that we have chosen $V = 0 \text{ V}$ to be the potential at $r = \infty$. So, we are obtaining potential and potential energy relative to a zero of potential and potential energy at infinity. Thus,

$$U_A = U_B = (-e)(V) = (-1.6022 \times 10^{-19} \text{ C})(1797.5 \text{ V}) = -2.88 \times 10^{-16} \text{ J}$$

$$U_C = (-1.6022 \times 10^{-19} \text{ C})(898.75 \text{ V}) = -1.44 \times 10^{-16} \text{ J}$$

- The potential differences are

$$\Delta V_{AB} = V_B - V_A = V_B - V_B = 0 \text{ V}$$

$$\begin{aligned} \Delta V_{BC} &= V_C - V_B \\ &= 898.75 \text{ V} - 1797.5 \text{ V} \\ &= -898.75 \text{ V} \end{aligned}$$

eV is energy!!

17

Electric Potential with Multiple Charges

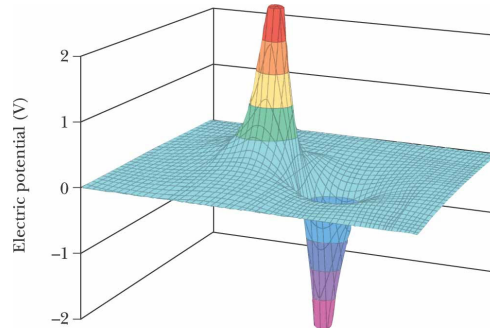
- The electric potential due to several point charges is the algebraic sum of the potentials due to each individual charge

$$V = k \sum_i \frac{q_i}{r_i}$$

This is an example of the superposition principle for scalar.

18

Electric Potential of a **Dipole** (positive charges and negative charges of the same quantity separated by a distance r)

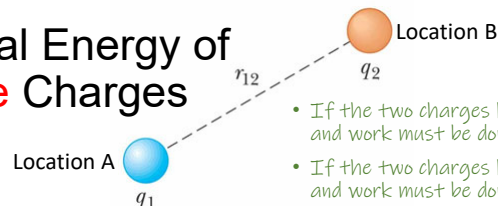


- Potential is 0 at very far away (plus and minus) infinity
- Potential is also 0 at the center of dipole.

(likewise for life !!!!)

19

Potential Energy of Multiple Charges



- If the two charges have the same sign, U is positive and work must be done to bring the charges together
- If the two charges have opposite signs, U is negative and work must be done to keep the charges apart

The potential energy of the system with q_1 and q_2 is computed as follows:

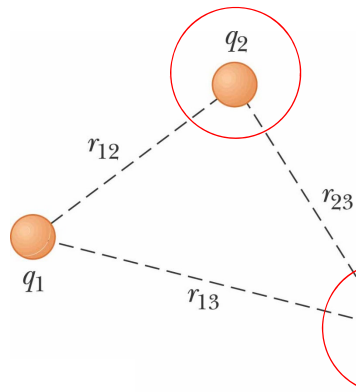
- We first move the two charges to infinity so that the field of the space enclosing A and B is 0 as there is no charge in the two location.
- Next we move charge q_1 to Location A in the free space, and the work required is 0 because the electric field is 0 in free space.
- Subsequently we move charge q_2 to location B. The **work required** on q_2 is computed as follows:

$$V = \frac{U}{q_2} \rightarrow U = q_2 \times V = q_2 \times \left(k \frac{q_1}{r_{12}} \right) = k \frac{q_1 q_2}{r_{12}}$$

- So the potential energy of the system with q_1 and q_2 $= 0 + k \frac{q_1 q_2}{r_{12}} = k \frac{q_1 q_2}{r_{12}}$ The potential energy can be (+) or (-).

20

Potential Energy with Multiple Charges, final



- If there are more than two charges, then compute U for each pair of charges and add them

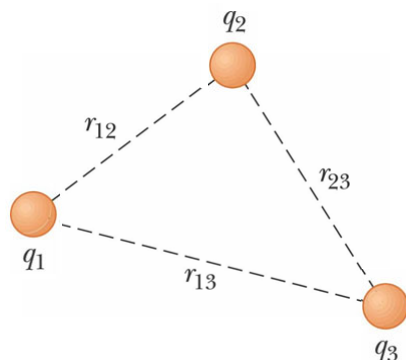
E.g., for three charges:

$$U = k \left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}} \right)$$

This result is independent of the order we bring in the charges

21

Potential Energy with Multiple Charges, final



$$U = k \left(0 + \frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}} \right)$$

$$U = k \left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}} \right)$$

22

Finding Electric Field **E** from Electric Potential Gradient

- Assume the direction of the **electric field E** has only an x component, then

$$E_x = -\frac{dV}{dx}$$

- Similar statements would apply to the y and z components:

$$E_y = -\frac{dV}{dy}, \text{ and } E_z = -\frac{dV}{dz}$$

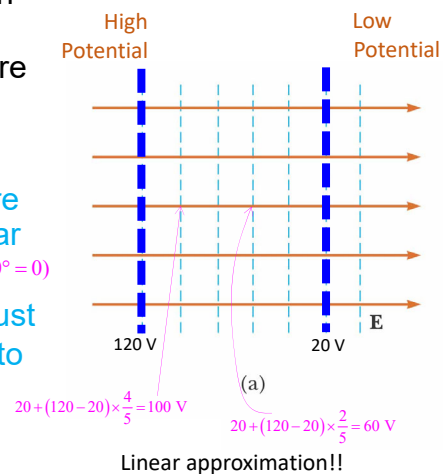
(In this course we will keep the potential to one dimension only.)

23

Equipotential Surface

E and **V** for an Infinite Sheet of Charge

- The electric field lines are represented by the brown lines in the figure
- The **equipotential lines** are represented by the **blue dashed lines**
- The equipotential lines are every where perpendicular to the field lines (so that $\cos 90^\circ = 0$)
- Equipotential surfaces must always be perpendicular to the electric field lines passing through them

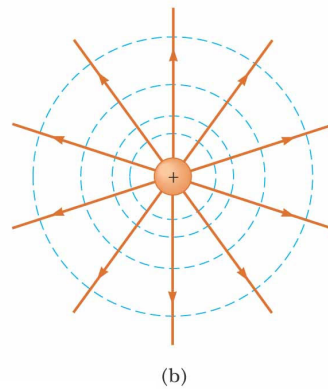


24

Equipotential Surface

$\mathbf{E}$ and V for a Point Charge

- The electric field lines are the brown lines
- The equipotential lines are the dashed blue lines
- The equipotential lines are everywhere perpendicular to the field lines

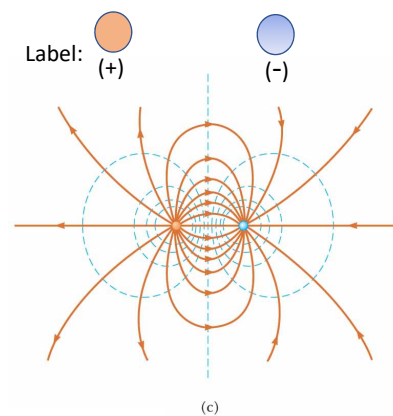


25

Equipotential Surface

$\mathbf{E}$ and V for a Dipole

- The electric field lines are the brown lines
- The equipotential lines are the dashed blue lines
- The equipotential lines are everywhere perpendicular to the field lines



26

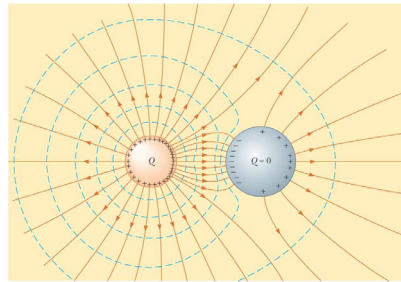
Irregularly Shaped Objects, cont.

(charges can freely move around)

For a solid conductor, all charges will be pushed to the surface. Inside the solid conductor the electric field is 0 everywhere.

Why?

- The field lines are still perpendicular to the conducting surface at all points
- The equipotential surfaces are perpendicular to the field lines everywhere

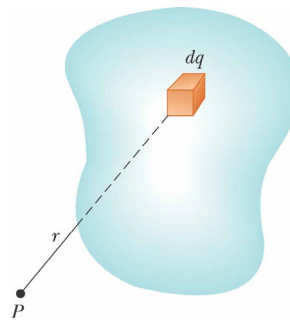


27

Electric Potential for a Continuous Charge Distribution

- Consider a small charge element dq
 - Treat it as a point charge
- The potential at some point P due to this charge element is

$$dV = k \frac{dq}{r}$$

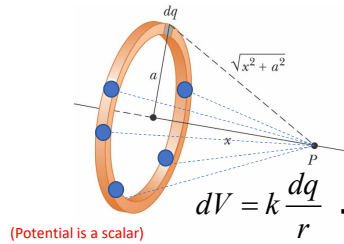


- To find the total potential, we need to integrate to include the contributions from all the small elements
- This value for V uses the reference of $V = 0$ when point P is infinitely very far away from the charge distributions

$$V = k \int \frac{dq}{r}$$

28

Electric Potential at Point P from a Uniformly Charged Ring



- P is located on the perpendicular central axis of the uniformly charged ring
- The ring has a radius a and a total charge Q

$$dV = k \frac{dq}{r} \rightarrow V = k \int \frac{dq}{r} = k \int \frac{dq}{\sqrt{x^2 + a^2}}$$

But any dq on the ring is at the same distance from point P . So,

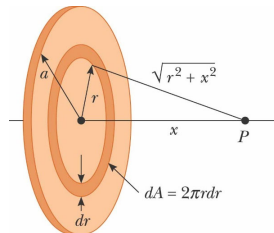
$$V = \frac{k}{\sqrt{x^2 + a^2}} \int dq = \boxed{\frac{kQ}{\sqrt{x^2 + a^2}}} \quad x \geq 0$$

Electric field at point P :

$$E_x = -\frac{dV}{dx} = -kQ \frac{d}{dx} (x^2 + a^2)^{-1/2} = -kQ \left(-\frac{1}{2} \right) (x^2 + a^2)^{-3/2} (2x)$$

$$= \boxed{\frac{kQx}{(x^2 + a^2)^{3/2}}} \quad \text{Same as Lecture-16} \quad x \geq 0$$

29



Electric Potential at Point P of Distance x from a Uniformly Charged Disk with Surface Charge Density σ

$$V_{\text{ring}} = \frac{kq}{\sqrt{x^2 + a^2}}$$

We can simplify the problem by dividing the disk into a series of concentric charged rings of infinitesimal width dr . The electric potential due to each ring is given by the previous answer. Consider one such ring of radius r and width dr , the charges in the ring is $dq = \sigma \times dA = \sigma \times (2\pi r \times dr)$.

$$dV = \frac{k(dq)}{\sqrt{r^2 + x^2}} = \frac{k(\sigma \times (2\pi r \times dr))}{\sqrt{r^2 + x^2}}$$

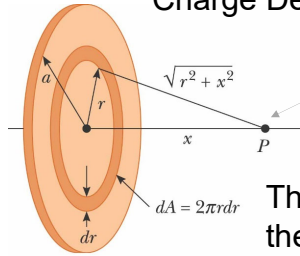
To find the **total electric potential** at P due to the disk, we sum over all rings making up the disk. That is, we integrate (**accumulate**) dV from $r = 0$ to $r = a$.

$$V = \pi k \sigma \int_0^a \frac{2r dr}{\sqrt{r^2 + x^2}} = \pi k \sigma \int_0^a (r^2 + x^2)^{-1/2} 2r dr = 2\pi k \sigma \int_0^a \left(\frac{1}{2} (r^2 + x^2)^{-1/2} 2r \right) dr$$

$$= 2\pi k \sigma \left[(r^2 + x^2)^{1/2} \right]_0^a = \boxed{2\pi k \sigma \left[(x^2 + a^2)^{1/2} - x \right]} \quad x \geq 0$$

30

Electric Field at Point P of Axial Distance x from a Uniformly Charged **Disk** with Surface Charge Density σ



$$V = 2\pi k\sigma \left[(x^2 + a^2)^{\frac{1}{2}} - x \right] \quad x \geq 0$$

The electric field at any axial point of the disk can be derived as follows:

$$E_x = -\frac{dV}{dx} = -\frac{d \left(2\pi k\sigma \left[(x^2 + a^2)^{\frac{1}{2}} - x \right] \right)}{dx}$$

$$= -2\pi k\sigma \left(\frac{1}{2} (x^2 + a^2)^{-\frac{1}{2}} (2x) - 1 \right)$$

$$= 2\pi k\sigma \left(1 - \frac{x}{\sqrt{x^2 + a^2}} \right) \quad (\text{for } x > 0)$$

Same as Lecture-16 (replaced a by R)

31

Example. An **insulating** solid sphere of radius R has a uniform positive volume charge density and total charge Q . Find the electric potential at a point outside the sphere, that is, $r > R$. Take the potential to be zero at $r = \infty$.

Answer:

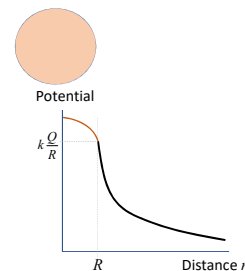
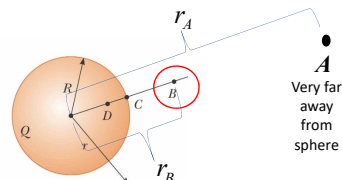
The electric field **outside** an insulating solid sphere can be treated as the field from a point charge where $r > R$. Thus to obtain the electric potential at an exterior point B , we let the reference point A to be at infinity as follows:

$$V_B - V_A = kQ \left[\frac{1}{r_B} - \frac{1}{r_A} \right] \rightarrow \infty$$

$$\Rightarrow V_B - 0 = kQ \left[\frac{1}{r_B} - 0 \right] \Rightarrow V_B = k \frac{Q}{r} \quad (\text{for } r > R)$$

Because the potential must be continuous at $r = R$, we can use this expression to obtain the potential at the surface of the sphere. That is, the potential at a point

such as C shown in the figure is $V_C = k \frac{Q}{R}$



32

Charge Re-distribution on Conductors

Example. Two spherical **solid conductors** of radii r_1 and r_2 are separated by a distance much greater than the radius of either sphere. The spheres are connected by a very thin conducting wire as shown in figure. The charges on the spheres in equilibrium are q_1 and q_2 respectively, and they are uniformly charged. Find the **ratio of the magnitudes of the electric fields** at the surfaces of the spheres.

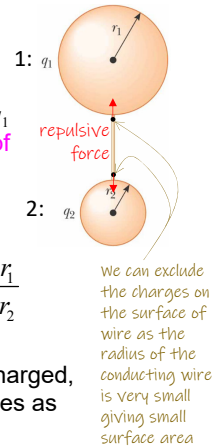
Answer:

Because the spheres are connected by a conducting wire, they must both be at the same electric potential when electrostatic equilibrium is reached.

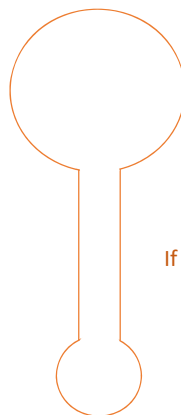
Because the spheres are far apart and their surfaces uniformly charged, we can express the magnitude of the **electric fields** at their surfaces as

$$\left. \begin{array}{l} E_1 = k \frac{q_1}{r_1^2} \text{ because } q_2 \text{ is very far away, and} \\ E_2 = k \frac{q_2}{r_2^2} \text{ because } q_1 \text{ is very far away.} \end{array} \right\} \rightarrow \frac{E_1}{E_2} = \frac{q_1}{q_2} \times \frac{r_2^2}{r_1^2} = \frac{r_1}{r_2} \times \frac{r_2^2}{r_1^2} = \frac{r_2}{r_1}$$

Hence, the field is more intense in the vicinity of the smaller sphere even though the electric potentials of both spheres are the same.



33



If the wire is thick

Equipotential Surface

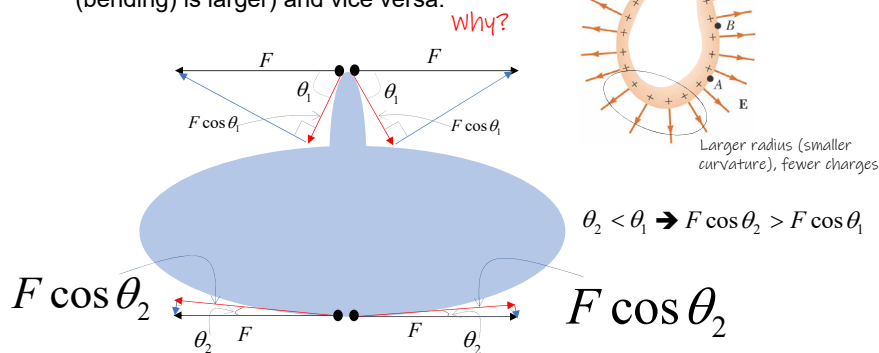
34

Forces on a Charged Conductor

- Charges can flow freely on a conductor which can be arbitrarily shaped. Thus **all charges are pushed to the surface by the electrostatic force until equilibrium is reached.**
- When the electrostatic equilibrium is reached, the charges are **not** distributed uniformly on the surface. The surface will have more charges if the radius there is smaller (or curvature (bending) is larger) and vice versa.

$$\text{curvature} = \frac{1}{r}$$

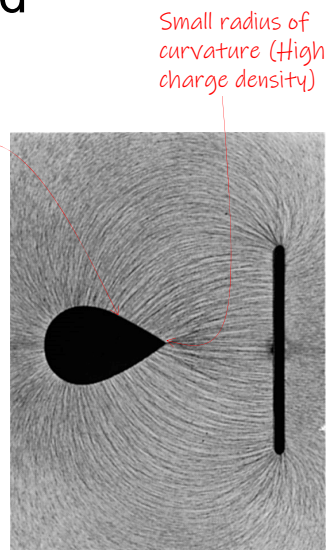
The curvature at a point tells how fast the curve is bending at that point!!



35

Irregularly Shaped Objects

- The charge density is high at the sharp end where the radius of curvature is small (and low where the radius of curvature is large)
- The electric field is large near the convex points having small radii of curvature and reaches very high values at sharp points



Curvature is the reciprocal of radius.

36

Potential Due to a Charged Solid Conductor

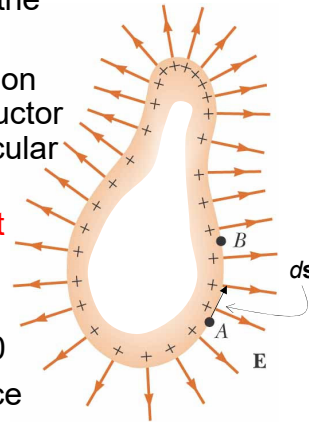
- During the electrostatic equilibrium, the field inside the conductor is 0 as the net force at any point inside the conductor is 0.
- Consider two points from A to B on the surface of the charged conductor as shown, $\mathbf{E}$ is always perpendicular to the small displacement $d\mathbf{s}$
- The work required to move a **unit** charge for a small displacement $d\mathbf{s}$ is

$$E = \frac{F}{q}$$

1 C

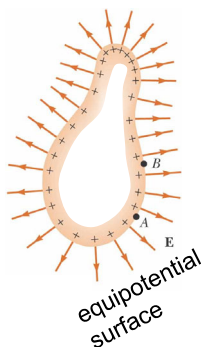
$$\mathbf{F} \cdot d\mathbf{s} = |\mathbf{E}| \times 1 \times |d\mathbf{s}| \times \cos 90^\circ = 0$$

- Therefore, the potential difference between A and B is also zero



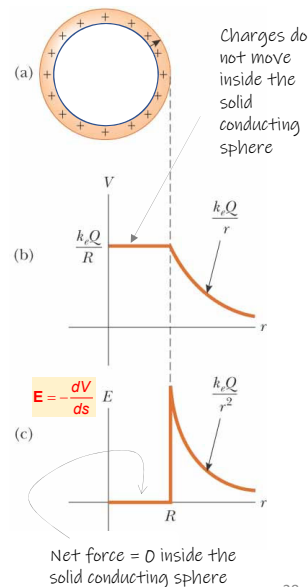
37

Potential Due to a Charged Conductor (Not Hollow), cont.



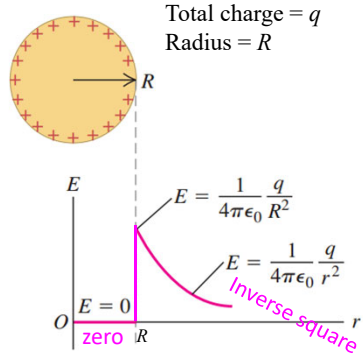
- So we conclude that in electrostatic equilibrium, potential is constant everywhere on the surface of a charged conductor, i.e., $\Delta V = 0$ between any two points A and B on the surface
- The **surface** of any charged conductor in electrostatic equilibrium is an **equipotential surface** (so it doesn't require any energy to move around the surface)

- Because the electric field is zero inside the solid conductor, we also conclude that the electric potential is constant everywhere inside the conductor and equal to the value at the surface



38

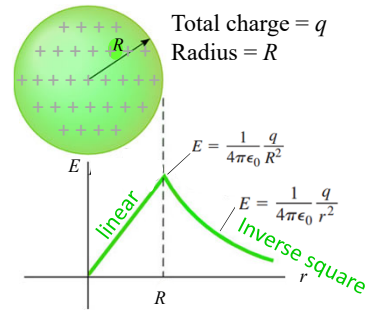
**Solid (not Hollow)
Conducting Sphere**



constant

hyperbolic

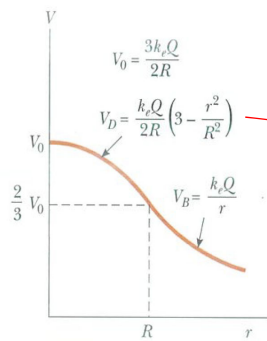
**Solid (not Hollow)
Insulating Sphere**



parabolic

hyperbolic

K

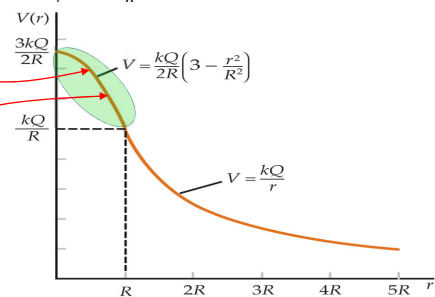
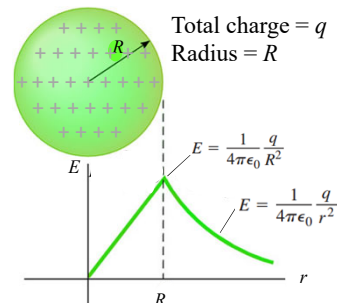


Inside an solid (not hollow) insulating sphere, the electric potential is

$$\text{constant}_1 \times \left(\text{constant}_2 - \frac{r^2}{\text{constant}_3} \right)$$

parabola!!

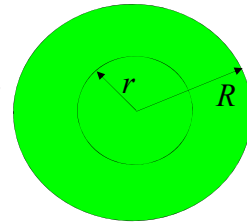
**Solid (not Hollow)
Insulating Sphere**



40

Potential inside a charged insulating sphere

Let the charge density be $\rho = \frac{Q}{\left(\frac{4}{3}\pi R^3\right)}$. From Gauss's Law: $\oint \mathbf{E} \cdot d\mathbf{A} = \frac{Q}{\epsilon_0}$



$$E \times (4\pi r^2) = \frac{\left(\frac{Q}{\frac{4}{3}\pi R^3}\right) \times \left(\frac{4}{3}\pi r^3\right)}{\epsilon_0} \rightarrow E(r) = \left(\frac{Q}{4\pi\epsilon_0 r^2 R^3}\right) r^3 = k \left(\frac{Q}{R^3}\right) r \quad (\text{linear})$$

From potential gradient, $E(r) = -\frac{dV}{dr} \rightarrow \int_r^R dV = -\int_r^R E(r') dr'$

$$\rightarrow V(R) - V(r) = -\int_r^R k \left(\frac{Q}{R^3}\right) r' dr' = -k \frac{Q}{R^3} \left[\frac{r'^2}{2}\right]_r^R = -k \frac{Q}{R^3} \left(\frac{R^2}{2} - \frac{r^2}{2}\right)$$

Since $V(\infty) = 0$, $V(R) = k \frac{Q}{R} \rightarrow k \frac{Q}{R} - V(r) = -k \frac{Q}{R^3} \left(\frac{R^2}{2} - \frac{r^2}{2}\right)$

41

Potential inside a charged insulating sphere (cont.)

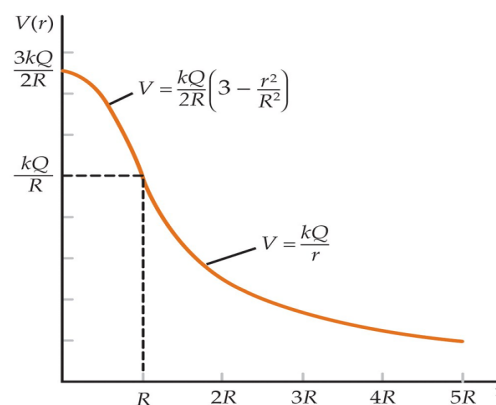
$$k \frac{Q}{R} - V(r) = -k \frac{Q}{R^3} \left(\frac{R^2}{2} - \frac{r^2}{2}\right)$$

$$V(r) = k \frac{Q}{R} + k \frac{Q}{R^3} \left(\frac{R^2}{2} - \frac{r^2}{2}\right)$$

$$= 2k \frac{Q}{2R} + k \frac{Q}{2R} - k \frac{Qr^2}{2(R \times R^2)}$$

$$= \frac{kQ}{2R} \left(3 - \frac{r^2}{R^2}\right)$$

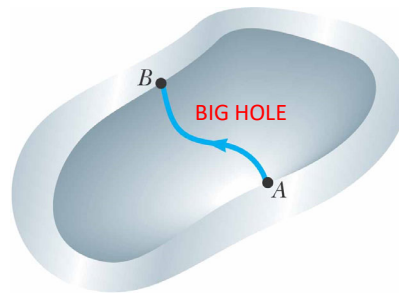
(parabolic)



42

Cavity in a Conductor (Big Hole Inside)

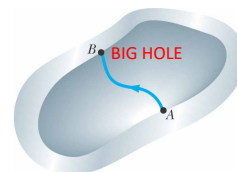
- Assume an irregularly shaped cavity is inside a conductor
- As no charges are inside the cavity, the electric field inside the conductor is zero



43

Cavity in a Conductor, cont

$$\mathbf{E} = -\frac{dV}{ds}$$



- The electric field inside does not depend on the charge distribution on the outside surface of the conductor
- For all paths between A and B *inside the cavity*,

$$V_B - V_A = -\int_A^B \mathbf{E} \cdot d\mathbf{s} = 0$$

- A cavity surrounded by conducting walls is a field-free region as long as no charges are placed inside the cavity

44



Photo Credit: <https://www.dreamstime.com/umbrella-raindrops-wet-green-leaves-rain-rainy-season-overcast-close-up-view-to-umbrella-umbrella-under-tree-image184784339>

Why you should not hold an umbrella under a tree when it is raining?

45

Observation

Corona Discharge



Lightning is the result of the build up of electrostatic charge in clouds. Positive and negative charges separate. Negative charges usually go towards the bottom of the cloud, while positive charges go to the top. After a certain amount of time, the negative charge leaps, connecting with either another cloud or even the ground.

- If the electric field near a conductor is sufficiently strong, electrons resulting from random ionizations of air molecules near the conductor accelerate away from their parent molecules
- These electrons can ionize additional molecules near the conductor thus more free electrons are created. The **corona discharge** is the glow that results from the recombination of these free electrons with the ionized air molecules. The ionization and corona discharge are most likely to occur near very sharp points.



46



Lightning bolt striking an HDB block in Strathmore Avenue at 2.55pm on Apr 23, 2022.
(Photo credit: Mak Wei Seng)

47



Photo Credit: <https://www.dreamstime.com/umbrella-raindrops-wet-green-leaves-rain-rainy-season-overcast-close-up-view-to-umbrella-umbrella-under-tree-image184784339>

Why you should not hold an umbrella under a tree when it is raining?

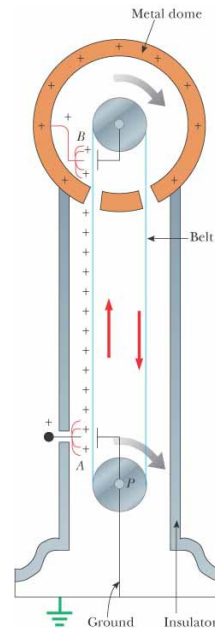
As the tip of the umbrella is a conductor, the corona discharge from the cloud can be directed to the tip of the umbrella. So the electron beam can strike the tree trunk and the trunk can fall on surrounding persons, or the electron beam can strike on the person directly.

48

Application

Van de Graaff Generator

- Belt is charged at point *A* by corona discharge from metallic needles.
- These charges are delivered continuously by moving belt of insulating material to a high-potential electrode at point *B*.
- The high-voltage electrode is connected to a hollow metallic dome mounted on an insulated column
- Large potentials on the dome can be developed by repeated trips of the belt
- Such large potentials can deliver enough energy to accelerate protons for the initiation of nuclear reactions
- The energy comes from the motor



Application

Electrostatic Shielding

- The cage is made of conducting wires supported on insulators.
- It is placed near an induction machine (such as the Van de Graaff Generator) that generates high potential.
- When the high voltage generates the corona discharge in the proximity of the cage, the people inside the cage is safe, and the electroscope there has no deflection because the electric field inside the cage is 0.



Faraday Cage



Singapore science centre faraday cage

51

Air pollution by coal-fired power plants



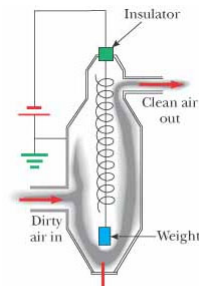
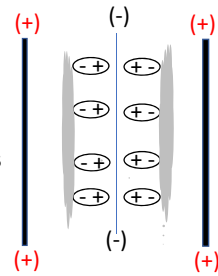
What can you do?

52

Application

Electrostatic Precipitator

- An application of electrical discharge in gases is the electrostatic precipitator
- It removes particulate matter from combustible gases – thus reducing air pollution
- The air to be cleaned enters the duct and moves near the wire
- As the electrons and negative ions created by the discharge are accelerated toward the outer wall by the electric field, the dirt particles become charged
- Most of the dirt particles are negatively charged and are drawn to the walls by **the electric field**.
- When the duct is periodically shaken, the particles break loose and are collected at the bottom.



53

Air pollution by power plants



Please put your knowledge to good use to mitigate the accumulation of carbon footprint!

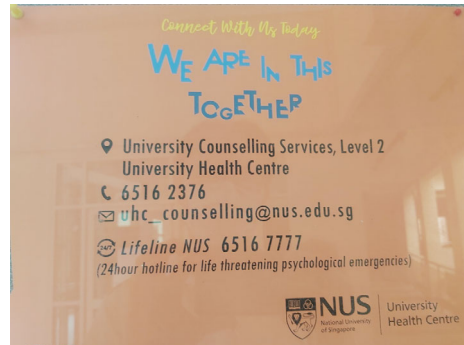
54

Helplines to de-stress

We are not superman/superlady. Sometime and somewhere in our life journey we will need help. Here they are:

Lifeline NUS (for life threatening psychological emergencies):
6516 7777 (24 hours)

6516 2376 (Office hours)
uhc_counselling@nus.edu.sg



Samaritans of Singapore Hotline: 1800 221 4444
Institute of Mental Health's Helpline: 6389 2222
Singapore Association of Mental Health Helpline: 1800 283 7019

55

- End -

56